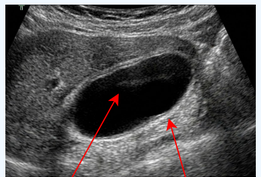
1 INPUT

$I \in \mathbb{R}^{H \times W}$



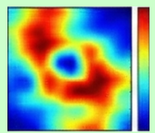
Speckle-Corrupted Image

2 SRAD FILTER



Anisotropic Diffusion

$$c(q) = \frac{1}{1 + \frac{q^2 - q_0^2}{q_0^2(1+q_0^2)}}$$



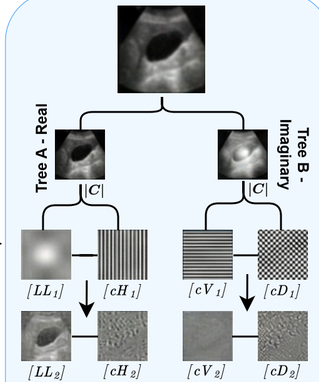
$\text{iterations} = 15,$
$\Delta t = 0.05,$
$\kappa = 30$



I_{SRAD}

Normalization

3 DT-CWT DECOMPOSITION



$C = \text{Tree}_A + j \cdot \text{Tree}_B$

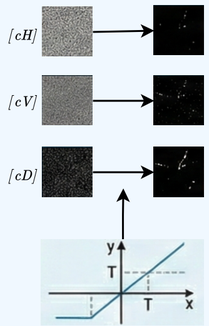
$|C|$ = Magnitude

$|C|$

Wavelet: Daubechies-4 (db4), Levels: 2

Noise Estimation

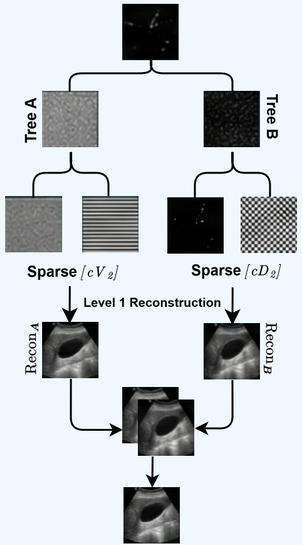
4 BAYESSHRINK THRESHOLDING



Soft Threshold Function

$C_{thresh} = \max(|C| - T_k,\ 0) \cdot e^{j\phi}$

5 INVERSE DT-CWT



$I_{denoised} = \frac{\text{Recon}_A + \text{Recon}_B}{2}$

De-Normalization

Scale and Clip

6 OUTPUT

